

# Esophageal Cancer Treatment

## Esophageal Cancer Treatment Categories

| Category             | AJCC | Stage          | Treatment                                               |
|----------------------|------|----------------|---------------------------------------------------------|
| High-grade Dysplasia | 0    | Tis            | Radiofrequency Ablation                                 |
| Superficial Tumors   | I    | T1a            | Endoscopic Therapy                                      |
| Localized Tumors     | IIB  | T1b T2         | Surgery                                                 |
| Locally-advanced     | III  | T3 or          | Chemo $\pm$ RT $\rightarrow$                            |
|                      | IIA  | N <sup>+</sup> | Surgery                                                 |
| Extra-Regional       | IVA  | N3             | Chemo $\rightarrow$<br>ChemoRT $\rightarrow$<br>Surgery |
| Metastatic           | IVB  | M1             | Chemotherapy<br>$\pm$ Radiation                         |

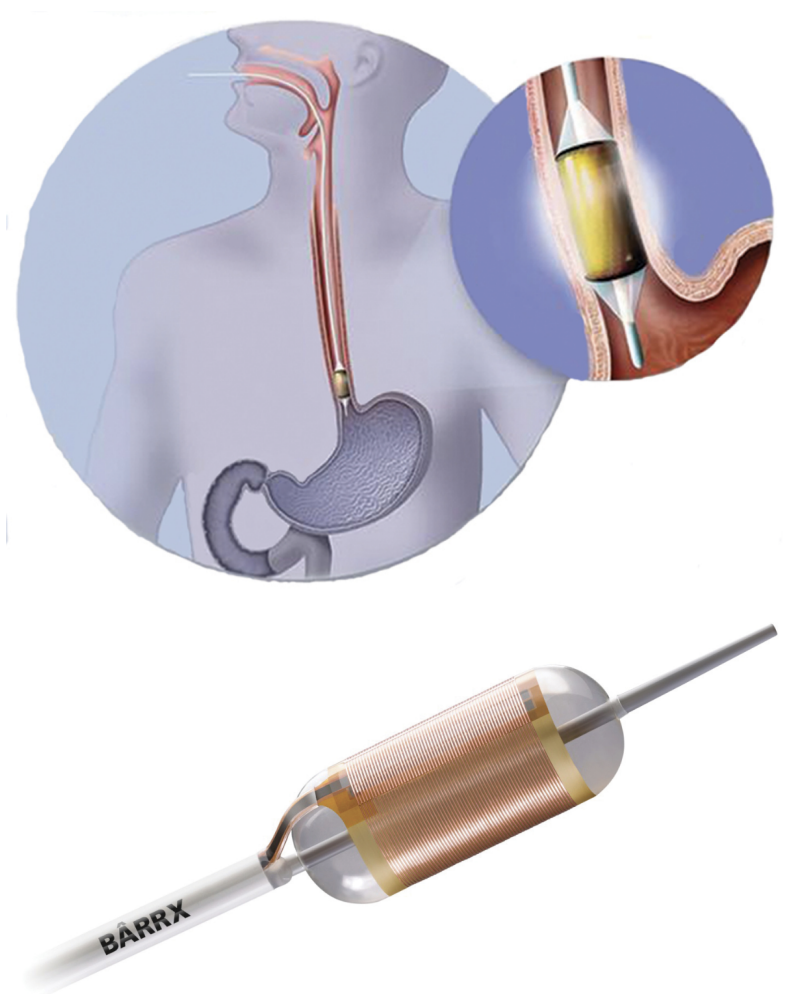
## High-grade Dysplasia $\Rightarrow$ Radiofrequency Ablation

127 patients with dysplasia randomized:

- Radio-frequency ablation
- Sham ablation

Low-grade dysplasia in 64

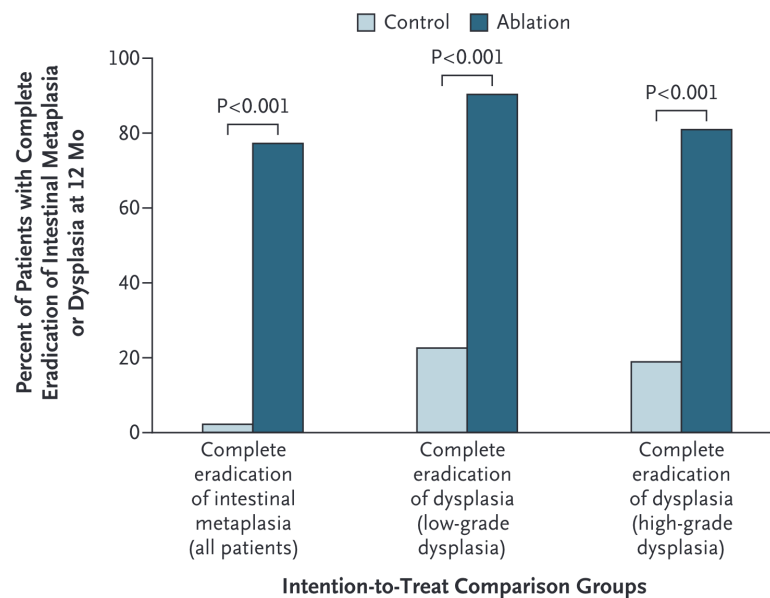
High-grade dysplasia in 63



(Shaheen et al. 2009)

## **Radiofrequency Ablation for Dysplasia**

Radiofrequency Ablation results in eradication of Barrett's in 75% at 1 year



No role for primary surgery in high-grade dysplasia

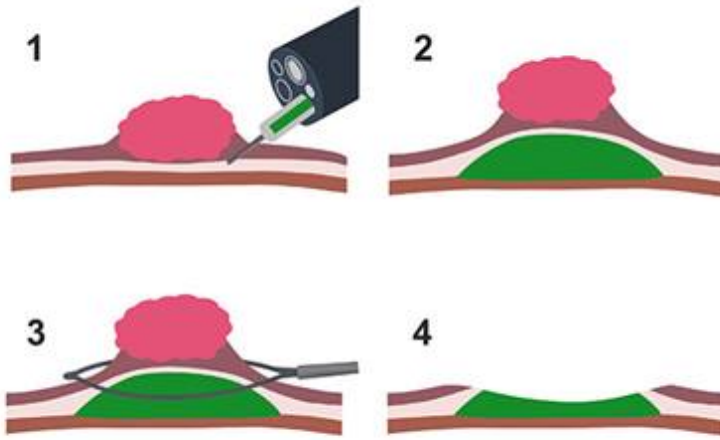
(Shaheen et al. 2009)

## Superficial Tumors (T1)

Workup of nodular Barretts:

- Endoscopic Ultrasound
- Endoscopic Mucosal Resection
  - Diagnostic (T staging)
  - May be therapeutic for T1a tumors

## Endoscopic Musocal Resection for T1 Tumors



## Endoscopic Submucosal Dissection for T1a/T1b

Endoscopic resection uses needle-knife to dissect *below* submucosa

Maybe suitable for T1b lesions *if* lesion is completely resected

Risk of perforation

## Localized Tumors (T2 N0)

Patients staged as uT2 N0 are candidates for primary surgery.  
*However:*

- EUS has a 25% rate of understaging uT2 N0 tumors
- Understaged patients who undergo primary surgery would need chemo or chemoRT postop

## Asymptomatic Tumors (minimal dysphagia)

- EUS to distinguish T2 from T3 tumors
- If uT2 N0 → CT chest/abdomen/pelvis → Esophagectomy
- If uT3 or N1 → PET → neoadjuvant therapy

Patients with dysphagia almost always are T3 tumors (and don't need EUS)

## Symptomatic Tumors (dysphagia)

Patients with dysphagia to solids or weight loss or tumor length >3cm are unlikely to have T1-2 tumors and can be initially evaluated with [PET Scan](#)

- Disease confined to the esophagus and regional nodes → [Locally-advanced](#)
- Metastatic disease → [Metastatic](#)
- N3 → [Extra-Regional](#)

## EUS in Patients with Dysphagia

Memorial Sloan Kettering patients with esophageal cancer:

- 61 with dysphagia, 54 (89%) were found on EUS to have uT3-4 tumors.
- 53 without dysphagia, 25 (47%) were uT1-2 → candidates for primary surgery.

EUS can be omitted for patients with dysphagia

EUS can be useful in patients *without* dysphagia (to identify T2N0 → Surgery)

(Ripley et al. 2016)

## PET Scan

PET has more specificity and sensitivity than CT in detecting regional lymph node and distal metastasis

## Locally-advanced

Improved survival with adjunctive therapy. There are two options:

- ChemoRT → Surgery ([CROSS Trial](#))
- Chemo → Surgery → Chemo ([EsoPEC Trial](#))

## CROSS Surgery vs ChemoRT → Surgery

368 esophageal cancer patients randomized:

- Surgery alone
- Chemo+RT → Surgery

75% adenocarcinoma. T3: 80% T2: 17%. age  $\tilde{x}$ =60

⇒ longer survival with Chemo+RT → Surgery

Chemotherapy: Weekly carboplatin and paclitaxel

Radiation: 4140 cGy in 23 fractions (180cGy/fraction)

(Shapiro et al. 2015)

## CROSS Surgery vs ChemoRT → Surgery

|                 | Surgeruy | CRT→Surgery |         |
|-----------------|----------|-------------|---------|
| Time to Surgery | 24d      | 97d         |         |
| Unresectable    | 13%      | 4%          | p=0.002 |
| R0 Resection    | 92%      | 69%         | p<0.001 |
| Positive Nodes  | 75%      | 31%         |         |
| Complications   | ~        | ~           |         |
| Mortality       | ~        | ~           |         |

## CROSS - Overall Survival

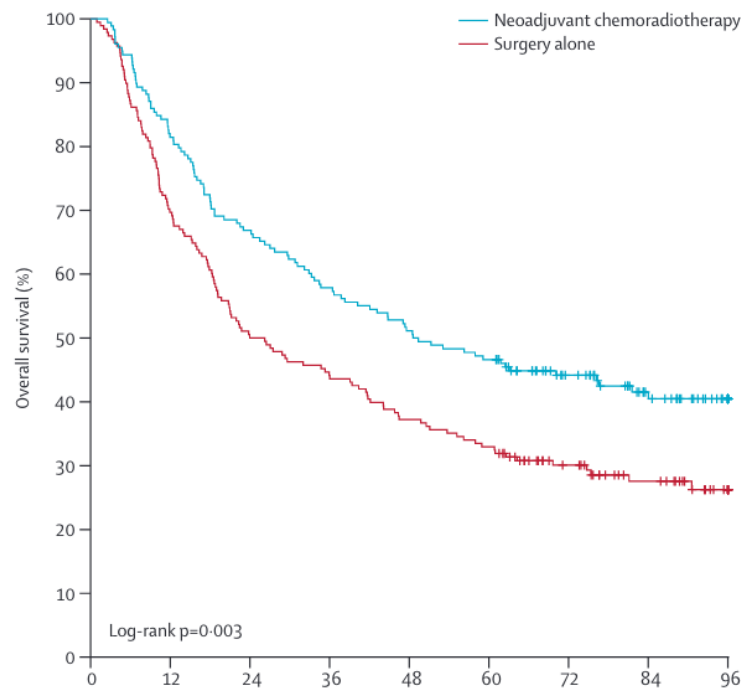


Figure 1: Surgery vs ChemoRT → Surgery

(Shapiro et al. 2015)

## CROSS - Survival by Histology

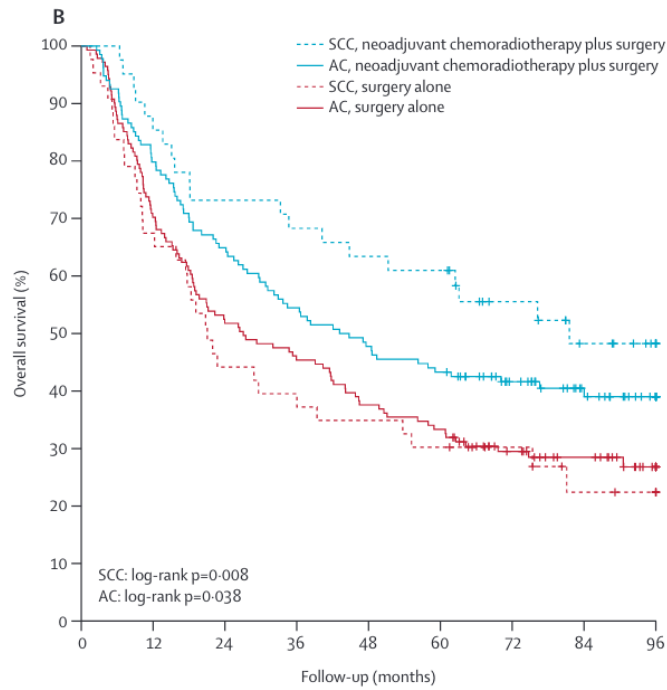


Figure 2: Surgery vs ChemoRT → Surgery

## CROSS Effects by Histology

(Shapiro et al. 2015)

### Adenocarcinoma

- Median survival 43mo vs 27mo
- Pathologic complete response in 23%

### Squamous Cell Carcinoma

- Median survival 82mo vs 21mo
- Pathologic complete response in 49%

### Surgical Therapy of Squamous Cell

(Shapiro et al. 2015)



## Adjuvant Nivolumab after CROSS: Checkmate 577 Trial

EsoCA patients who received ChemoRT → Surgery with residual disease (not pCR)

Randomized to one year of nivolumab vs Observation

⇒ Adjuvant nivolumab group had longer median survival: 22mo vs 11mo

(Kelly et al. 2021)

## Checkmate 577 Trial

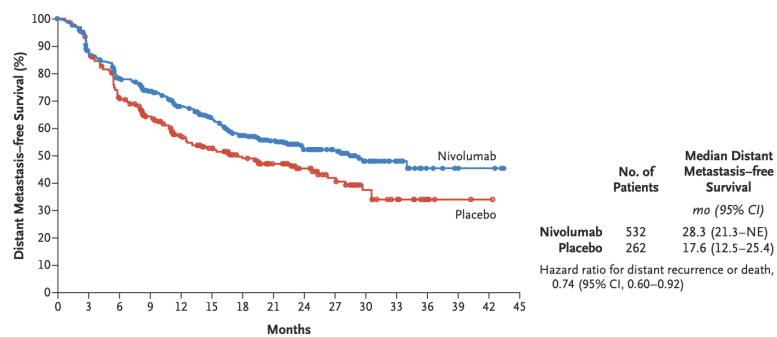


Figure 3: Adjuvant Nivolumab vs Observation

## Nivolumab = Immune Checkpoint Inhibitor (ICI)

(Kelly et al. 2021)

PD-L1 agonist ligand

Interferes with tumor cell down-regulation of T cells

Active against stage IV esophageal cancer

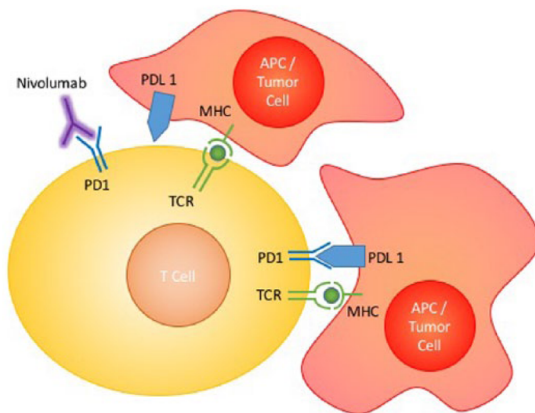


Figure 4: Nivolumab mechanism of action

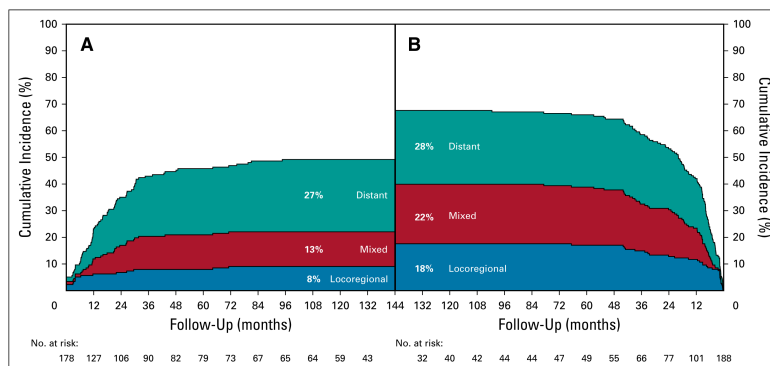
## CROSS - ChemoRT Does Not Reduce Distant Failure

Sites of failure over time

ChemoRT + Surgery *vs* Surgery

ChemoRT appears to reduce risk of local or local+distant failure, but not isolated distant failure

⇒ Can chemotherapy reduce risk of distant failure?



(Shapiro et al. 2015)

## EsoPEC CROSS vs FLOT for Adenocarcinoma

Adenocarcinoma esophagus and GE junction

- T1 N+ or T2-4a M0. Median age =63. 89% men

Randomized to CROSS (n=217) vs FLOT chemotherapy (n=221)

- CROSS: carboplatin/paclitaxel + 4140cGy → Surgery
- FLOT: FLOT → Surgery → FLOT

Excluded: Squamous cell, gastric cancer, T1N0, T4b, M1

(Hoeppner et al. 2025)

### EsoPEC: FLOT superior to CROSS

|                      | FLOT<br>Periop<br>Chemo | CROSS<br>ChemoRT | CROSS<br>Trial<br>ChemoRT |
|----------------------|-------------------------|------------------|---------------------------|
| 90d Mortality        | 3.2%                    | 5.6%             |                           |
| $\tilde{x}$ Survival | 66mo                    | 37mo             | 43mo                      |
| 3year Survival       | 57%                     | 51%              | 58%                       |
| 5year Survival       | 51%                     | 29%              | 47%                       |
| path CR              | 17%                     | 10%              | 23%                       |

See also [Neoadjuvant Chemo](#)

(Hoeppner et al. 2025)

### Surgery for Squamous Cell Carcinoma

Squamous Cell Carcinoma of the esophagus

- responds well to chemo+RT
- more difficult to get a surgical margin on the airway
- additional benefit of surgery on top of chemoRT is uncertain

### **Squamous Cell: ChemoRT $\pm$ Surgery (FFCD)**

Squamous cell of the esophagus  $\rightarrow$  4500cGy RT + 2 cycles of cisplatin + 5FU

Patients with a clinical response were randomized:

- Surgery  $\Rightarrow$  2 year survival 34% Median 17.7mo
- 3 cycles of chemo + 2000 cGy RT  $\Rightarrow$  2 year survival 40% Median 19.3mo

$\Rightarrow$  No difference in overall survival

(Bedenne et al. 2007)

### **Squamous Cell: ChemoRT $\pm$ Surgery (German)**

Squamous cell of the esophagus randomized:

- 4000 cGy RT + Chemo  $\rightarrow$  Surgery. 64% 2-year PFS. Mortality 12.8%
- 6500cGy RT + Chemo: 41% 2-year PFS. Mortality 3.5%

$\Rightarrow$  No difference in overall survival

(Stahl et al. 2005)

### **Non-Operative Management**

#### **Extra-regional Disease = N3**

Patients with N3 disease are Stage IVA and have a poor prognosis.

Treatment consists of chemotherapy followed by radiation.

Patients with persistent disease in the esophagus (and no distant disease) would then be candidates for surgery

## Metastatic = IVB

FOLFOX is first-line systemic therapy for metastatic GI cancers

- Dose-limiting toxicity is frequently peripheral neuropathy

## Adjuvant Immunotherapy: Checkmate 577 Trial

Immunotherapy with nivolumab as adjuvant therapy after CROSS regimen for patients with residual disease

Stage II/II Esophageal or GE junction cancers Adenocarcinoma or squamous cell

ChemoRT → Surgery *with residual disease on pathology*

Treatment Group: Nivolumab every 2 weeks x 4 months → every month x 8 months

*vs*

Control Group: No adjuvant therapy

⇒ Better survival in group with adjuvant nivolumab

(Kelly et al. 2021)

## Neoadjuvant Chemo for EsoCA

- MAGIC trial (gastric): ECF<sup>1</sup>→Surgery→ECF *vs* Surgery
- **OEO2 Trial**: (esophageal) Chemo→Surgery→ Chemo *vs* Surgery
- FLOT (gastric): FLOT<sup>2</sup>→Surgery→ FLOT *vs* ECF→Surgery→ECF
- **EsoPEC**: (esophageal):FLOT→Surgery→FLOT *vs* ChemoRT→Surgery (CROSS)

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<sup>1</sup>Epirubicin, Cisplatin, 5FU

<sup>2</sup>5FU, Leuvocorin, Oxaliplatin, Decetaxol

## OE02 Clinical Trial

- 802 Esophageal adenocarcinoma and squamous cell
- Randomized to Chemo → Surgery → Chemo *vs* Surgery alone
- Chemotherapy with ECF (Epirubicin, Cisplatin, 5FU)
- 5-year survival 23% for chemo+surgery vs 17% for surgery (HR 0.84 p=0.03)

(Allum et al. 2009)

## Neo-Aegis Trial CROSS vs MAGIC/FLOT

- **Adenocarcinoma** T2-3 N0-3 M0 Tumor length <8cm
- ChemoRT arm: carboplatin + paclitaxel + 4140cGy
- Chemo arm: MAGIC (ECF) or FLOT (later in trial)
- No difference in overall survival
- R0 resection 96% with CROSS vs 82% with chemo
- pCR 12% with CROSS vs 4% with chemo

(reynolds1015?)

## Orientation Manual



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